

FINAL SEMINAR · OCTOBER 2026

Multi-Page Visually Rich Document Understanding

A survey of system design and an empirical attribution of failure

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Roadmap

01 Background

02 **Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding**

Lewei Xu, Yihao Ding, Zihan Xu, Siwen Luo, Daochang Liu, Wei Liu

EMNLP 2026, main conference

03 **Locating Failure in Multi-Page Visually Rich Document Understanding: An Empirical Attribution**

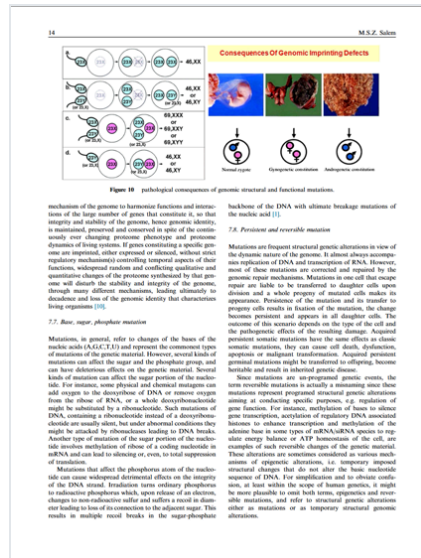
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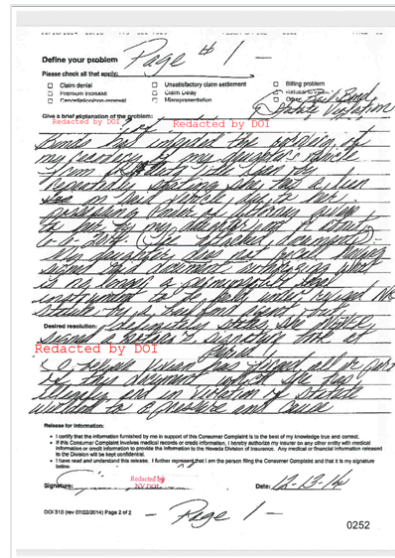
04 **Proof of concept: a simple method for document understanding**

Multi-Page Visually Rich Documents

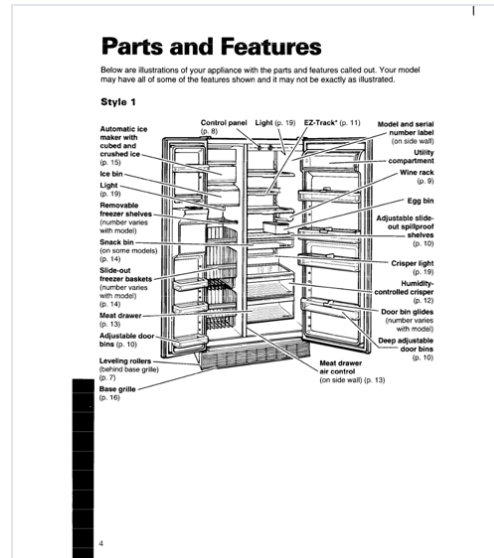
Meaning is carried jointly by **text, vision, and layout** rather than by text alone. **Multi-page** adds a further layer: meaning also runs between pages, as when a claim points to a figure many pages away.



Research paper
born-digital



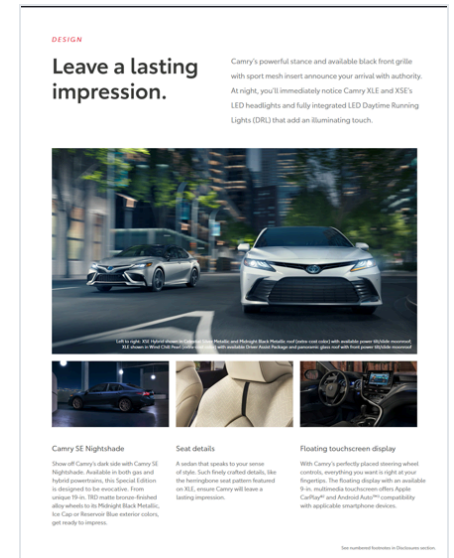
Signed form
scanned, handwritten



Technical manual
layout-dependent

Table C.10 (Contd.)						
Item C: Indefinite Intangible Financial Data						
The following reclassified consolidated financial data is not necessarily indicative of results of future operations and should be read in conjunction with Item 5 Management Discussion and Analysis of Financial Condition and Results of Operations and Item 6 Financial Statements and Supplementary Data. The following information is reclassified to restate prior year data and does not necessarily have been adjusted for the prior periods reported. See Note 1 and Note 2 to Item 6 Financial Statements and Supplementary Data for further detail on the Note Split.						
Consolidated Statements of Operations:						
	Year ended December 31,					
	2014	2014	2014	2013	2013	2013
	in thousands, except per share data					
Revenue	\$ 4,779,511	\$ 3,150,458	\$ 4,374,562	\$ 4,689,324	\$ 3,254,937	\$ 3,254,937
Operating expenses	390,428	462,440	224,147	49,992	17,566	17,566
Net income	322,641	266,799	112,403	17,512	17,512	278,213
Income per share:						
Basic	\$ 0.29	\$ 0.41	\$ 0.26	\$ 0.28	\$ 0.24	\$ 0.46
Diluted	\$ 0.28	\$ 0.40	\$ 0.26	\$ 0.28	\$ 0.24	\$ 0.43
Weighted average common shares outstanding:						
Basic	421,899	420,144	407,381	108,448	109,929	109,929
Diluted	486,436	479,899	423,327	412,327	401,187	401,187
Consolidated Statements of Cash Flows:						
	Year ended December 31,					
	2014	2014	2014	2013	2013	2013
	in thousands					
Net cash (at) provided by operating activities	\$ (149,433)	\$ 16,403	\$ 97,311	\$ 21,396	\$ 31,712	\$ 31,712
Free cash flow	(920,317)	(226,699)	(14,300)	(18,352)	186,510	186,510
Item D: "Liquidity and Capital Resources" for a definition of "free cash flow" and a reconciliation of "free cash flow" to net cash (at) provided by operating activities:						
Consolidated Balance Sheets:						
	as of December 31,					
	2014	2014	2014	2013	2013	2013
	in thousands					
Cash, cash equivalents and short-term investments	\$ 2,131,711	\$ 1,404,408	\$ 1,530,841	\$ 748,178	\$ 797,311	\$ 797,311
Current assets, net (1)	1,751,813	1,499,400	1,838,944	2,914,009	2,622,459	2,622,459
Working capital	1,862,706	1,380,009	1,610,695	3,168,187	2,859,386	2,859,386
Total assets (2)	10,262,311	7,942,500	5,464,623	1,964,761	1,662,111	1,662,111
Long-term debt (3)	2,173,342	1,807,489	494,162	167,762	192,593	192,593
Long-term debt as a related party (1)				188,109	197,782	197,782
Non-current intangible liabilities	2,606,340	3,575,652	1,345,590	1,676,622	1,799,628	1,799,628
Current liabilities	4,351,393	3,680,973	3,121,573	2,441,440	1,674,644	1,674,644
Total liabilities, equity	2,225,440	1,817,788	1,333,944	764,673	462,610	462,610
Total current year amounts have been reclassified to conform to the current year presentation in the consolidated financial statements. See Note 1 and Note 2 to Item 6 Financial Statements and Supplementary Data for further detail.						

Financial report
dense tables



Product brochure
photo-led

- **Text.** Small, typeset or handwritten, and legible only at far higher input resolution than a typical vision-language task requires.
- **Vision.** Charts, figures, photographs, and handwriting carry meaning that no text extraction recovers.
- **Layout.** Position is semantic, and reading order must be recovered rather than assumed.

What is Document Understanding?

One broad way to define it: get a system to **answer a natural-language question over a document**, accurately. The systems range from rule-based lookup, through trained vision-language models, to pipelines built around pretrained general-domain ones.

- | | | |
|---|---|-----------------------------|
| 1 | "What year was this report published?" | FACT LOOKUP |
| 2 | "According to the table, which region had the highest revenue in 2023?" | READ A TABLE |
| 3 | "How much did the leading region's revenue grow between the 2022 and 2023 results tables?" | TWO TABLES, THEN ARITHMETIC |
| 4 | "Does the growth described in the executive summary agree with the figures in the results table?" | PROSE AGAINST NUMBERS |
| 5 | "Across the report and its appendix, what evidence supports the conclusion that the new method improved performance, and what limitation qualifies it?" | GATHER, THEN QUALIFY |

Illustrative questions, not quoted benchmark items

LLM Assistants

Provide a demonstration of how you would use tool calls to answer a difficult question over this long visually rich document.

Designing a sample tool-based walkthrough for a cross-page question.

I'll demonstrate on a question that needs a figure and a table: "How often does the final pipeline stop at its earliest allowed point, and how does its page usage compare with the number of gold evidence pages?" First I'll map printed page numbers to PDF pages, then read the relevant pages as images.

Read a file, updated tasks ▾

Added task **Locate pages for Figure 4.2 and Table 4.2** ▾

Request

Description

Map printed page numbers to PDF pages using the contents list and the offset seen on page 27

Added task **Read figure and table pages visually** ▾

Request

Description Read PDF pages 60-61 and extract the bar heights and token statistics

Added task **Verify and answer** ▾

Request

Description Cross-check figure values against surrounding text, then answer

Completed task **Locate pages for Figure 4.2 and Table 4.2**

Started task **Read figure and table pages visually**

Read **Honours__Lewei_Xu.pdf** ▸

Verifying the data points plotted across the chart.

Completed task **Read figure and table pages visually**

Completed task **Verify and answer**

- A question over a 100-page thesis: the assistant plans, locates the pages, then answers.
- It renders the pages and reads them as images.

Read **Honours__Lewei_Xu.pdf** ▾

4. VALIDATION AND CONCLUSION

Method	OCR	Trained	Evidence source					Evidence pages				
			TEXT	LAT	CLS	YAS	FIG	SIN	MRG	UNA	AEC	
			Official protocol: answer extraction + rule-based scores									
Gold pages + VLM (VLM)	✓	✓	48.1	49.6	39.4	50.5	42.5	57.7	37.4	52.5	49.3	
Qwen2.5-72B	✓	✓	33.6	27.7	29.3	27.5	27.2	42.7	16.9	43.5	33.8	
LLaMA [14]	✓	✓	38.4	31.2	27.7	38.6	27.5	41.9	29.9	45.5	37.4	
MiniCPM [19], via [21]	✓	✓	43.1	31.2	32.8	38.7	37.9	35.5	23.8	28.1	38.5	
MiniCPM [19]	✓	✓	41.7	32.4	34.2	37.9	37.6	30.1	26.1	35.8	39.3	
MiniCPM [19]	✓	✓	42.7	38.8	33.3	38.5	40.7	32.9	27.8	35.9	41.0	
DocReader [12]	✓	✓	—	—	—	—	—	—	—	—	40.8	
Doc V [10]	✓	✓	—	—	—	—	—	—	—	—	42.1	
MiniCPM [19]	✓	✓	46.2	38.1	40.8	52.8	35.9	38.0	31.2	46.2	49.7	
LLM judge: binary correctness												
Gold pages + VLM (VLM)	✓	✓	55.3	50.9	45.5	48.4	44.8	63.5	38.6	51.1	52.2	
Qwen2.5-72B	✓	✓	—	—	—	—	—	—	—	—	31.9	
LLaMA [14]	✓	✓	43.2	40.0	36.1	31.6	42.0	31.5	24.8	8.7	35.5	
MiniCPM [19], via [21]	✓	✓	47.0	39.8	47.4	41.0	39.9	39.5	28.6	33.5	43.8	
MiniCPM [19]	✓	✓	49.7	42.0	44.6	37.8	42.1	38.7	31.7	62.1	50.4	
Qwen2.5-72B	✓	✓	—	—	—	—	—	—	—	—	38.8	
LLaMA [14]	✓	✓	52.7	42.4	46.6	36.3	45.3	42.2	39.3	75.4	56.3	
Doc V [10]	✓	✓	—	—	—	—	—	—	—	—	40.8	
Doc V [10]	✓	✓	—	—	—	—	—	—	—	—	40.8	

Table 4.1 MMLU accuracy (%) for published systems using open 72B vision-language backbones. Reported values are taken directly from the cited papers. "via" indicates results reproduced by another study rather than reported in the system's original paper. OCR, parser or OCR text is provided as input. Trained: task-specific fine-tuning or RL within the pipeline. A text LLM operates alongside the VLM. *Retriever only. *The document outline includes visual-element descriptions generated by Qwen2.5-VL-72B.

and layout questions decline. Multi-page accuracy also remains a clear weakness at 37.8%, below Doc-Trace's 41.3%, indicating that the higher overall accuracy does not eliminate the underlying difficulty of integrating evidence across pages.

Adaptive evidence acquisition. Figure 4.2 shows that the pipeline does not uniformly consume its full page budget. The largest group terminates after six pages, the earliest point permitted by the stopping floor $l = 6$, while progressively fewer questions continue deeper into the retrieval ranking and a smaller group reaches the $l = 25$ limit. Responed trajectories account for part of this longer tail, with some questions requiring one or multiple `CONTINUE` decisions after an initial synthesis attempt. This differs substantially from the annotated evidence distribution, where most answerable questions contain only one or two gold pages. The gap reflects the practical setting in which the locations of those pages are unknown and additional contextual pages may also contribute useful evidence.

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4. VALIDATION AND CONCLUSION

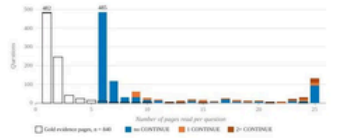


Figure 4.2 Trajectory of our method.

Turns	Tokens				Latency (s)		Peak
	Input text	Input visual	Output	Profile	Decoding	VRAM (GB)	
reader	11.2	5,209	834	1,008	2.19	70.6	7.3
synthesizer	1.4	9,638	1,742	724	4.0	59.0	8.2
retriever	1.3	4,088	0	1,083	1.6	72.8	7.1
Per question, mean	13.9	78,215	11,733	13,721	29.7	568.1	8.2
Per question, median	9.0	37,484	7,272	8,081	13.8	544.6	8.0

Table 4.2 Inference cost of the pipeline per agent turn and question.

Inference cost. Table 4.2 quantifies the cost of this adaptive behaviour. The pipeline uses 13.9 turns per question on average, compared with a median of 9.0, with the `reader` accounting for 11.2 turns and the `synthesizer` and `retriever` invoked only 1.4 and 1.3 times per question respectively. Mean peak memory is 8.2GB, remaining comfortably within the 16GB budget of the deployment GPU, while the average question consumes 78,215 text input tokens, 11,733 visual tokens, and 13,721 output tokens across its trajectory. The principal cost is latency rather than memory: mean profile time is only 29.7 s per question compared with 968 s of decoding, with a median total of roughly 9.3 minutes. The method therefore trades additional sequential inference for bounded memory use and broader evidence acquisition, with the deployment parameters providing a direct mechanism for reducing trajectory depth where lower latency is required.

4.2 Open Challenges and Future Work

The survey of Chapter 2 identifies five challenges that define the open MP-VREDU frontier. They arise from a consistent pattern across the surveyed systems: progress comes from moving evidence heading out of the backbone, through compression, retrieval, or iterative orchestration, but each shift relocates the bottleneck rather than removing it. Chapter 3 and the system-level validation in §4.1 address parts of the first three challenges directly, but also show that their underlying trade-offs remain unresolved. These concern system capability, in the evidence pipeline itself, in faithfulness, and in efficiency. Two concerns data and evaluation, in supervision integrity and in evaluation scope.

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Real-World Constraints

A hospital, a mining company, or a law firm wants the same thing: **classification, summarisation, or structured extraction over tens of thousands of documents**. The consumer path is often not available to them.

WHY NOT JUST CALL AN API

- **The documents cannot leave.** Medical records, legal and financial filings, and resources data carry privacy and data-residency requirements that rule out a third party.
- **The volume makes it uneconomic.** At tens of thousands of documents, per-call pricing stops being worth the benefit even where sending the data is permitted.
- **The document type is unusual.** A niche form or report may need its own training or fine-tuning rather than whatever a general model already knows.

WHAT THAT FORCES

- **Open-weight models, run locally.** A pretrained vision-language model on consumer hardware, not a datacentre.
- **A fixed compute budget.** Which pages are read, at what resolution, and how many times, all have to fit inside it.
- **Efficiency as a research problem.** Cost and latency are open questions in their own right, not implementation details.

This is where the research sits: **techniques and architectures for document understanding under a fixed compute budget**. The same techniques then flow back into the consumer products, which is why those have moved so quickly.

Survey of MP-VRDU

Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding

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Abstract

Documents spanning tens to hundreds of pages are pervasive in real-world workflows, from scientific papers to financial reports and technical manuals. In this setting, answer-relevant evidence is sparse and distributed across pages, and the document itself often exceeds a backbone MLLM's context window, making MP-VRDU a distinct problem from single-page VRDU. We survey MP-VRDU through the lens of evidence management, the organising principle that distinguishes multi-page from the single-page setting: the problem of deciding how document-scale evidence is selected and reasoned over when it cannot all be held at once. We introduce a taxonomy spanning four architectural families, multimodal representa-

MP-VRDU has seen rapid growth in recent years, with new architectures, training strategies, and inference methods emerging. Across these lines of work, the reasoning engine is almost always a general-domain MLLM; what varies is the evidence-management machinery built around it to handle documents whose length and structure exceed what the MLLM can process directly. Existing work has explored joint multi-page encoding (Tito et al., 2023), adaptation of pretrained MLLMs (Duan et al., 2025b), retrieval-augmented pipelines that decouple evidence selection from generation (Cho et al., 2024), and adaptive controllers that gather evidence iteratively (Sun et al., 2025). However, this body of work has developed along largely parallel lines, with limited cross-

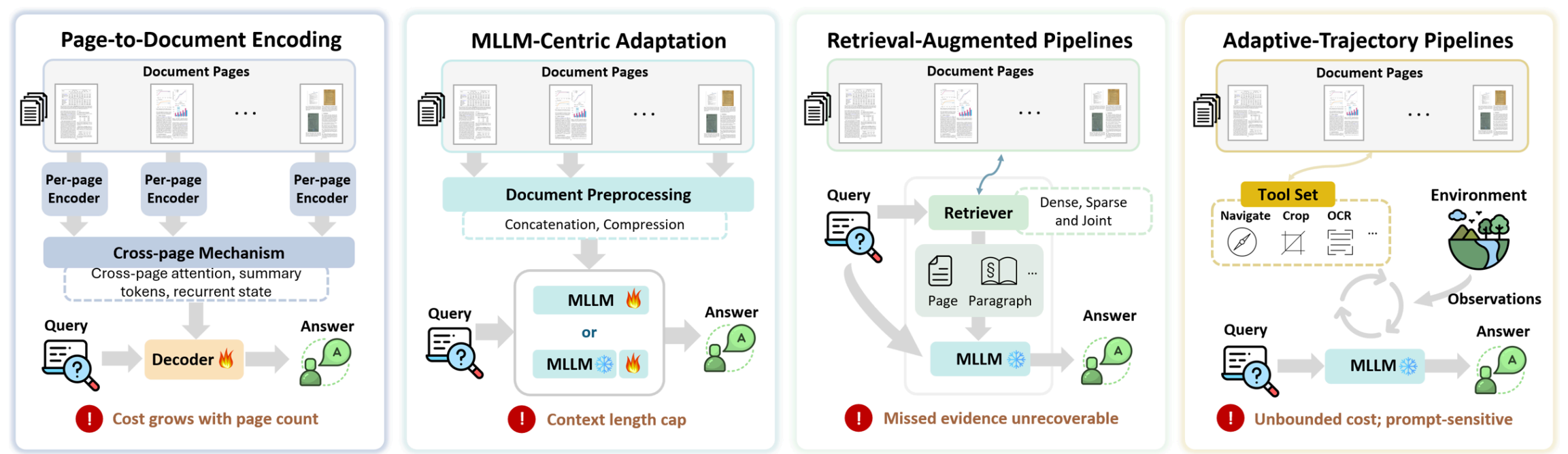
- 1 Introduction
- 2 Methodology and Scope
- 3 **MP-VRDU Architecture Overview**
- 4 Multi-Page Multimodal Representation
- 5 **Inference-Time Adaptation Strategies**
- 6 **Training Strategies**
- 7 Datasets
- 8 Challenges and Future Work

EMNLP 2026, main conference

Bold sections are covered in this talk

Multi-Page Architectural Overview

The survey organises systems by **evidence management**, into four architectural families. **Page-to-document encoding models** jointly encode pages; **MLLM-centric adaptation models** fit multi-page input into an MLLM’s context and token budget; **retrieval-augmented pipelines** retrieve evidence before generation; **adaptive-trajectory pipelines** inspect evidence iteratively under an LLM controller.



Snowflake indicates typically frozen, fire indicates typically trained

Inference-Time and Training Strategies

Three evidence-management decisions. Each can be **prompted** around a frozen **MLLM**, or **trained** into it.

Retrieval and navigation

TRAINING-FREE

Similarity-based or relation-aware retrieval

TRAINED

Modality-alignment, cross-page-fusion, logical-relevance

Reasoning strategies

TRAINING-FREE

Chain-of-Thought, Self-Reflection, Self-Consistency

TRAINED

Trace-imitation, then reward-driven (GRPO)

Agentic methods

TRAINING-FREE

ReAct and tool use, multi-agent orchestration

TRAINED

Trajectory-distillation, reinforcement-learned

Empirical Attribution Study

Locating Failure in Multi-Page Visually Rich Document Understanding: An Empirical Attribution

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Abstract

Multi-page visually-rich document understanding (MP-VRDU) requires managing evidence that is sparse, spread across pages, and often exceeds a model's context window. Prior work has produced competing, largely untested claims about how these systems should be built. We attribute incorrect answers to three failure modes, representation, selection, and reasoning, and isolate each over a multi-page document understanding dataset by intervening on one while holding the others fixed. We find that vision is necessary but does not replace text extraction, that missing pages bound accuracy while distractors cost little, and that reasoners fail to integrate evidence across pages even when it is fully supplied. Prompting can shift

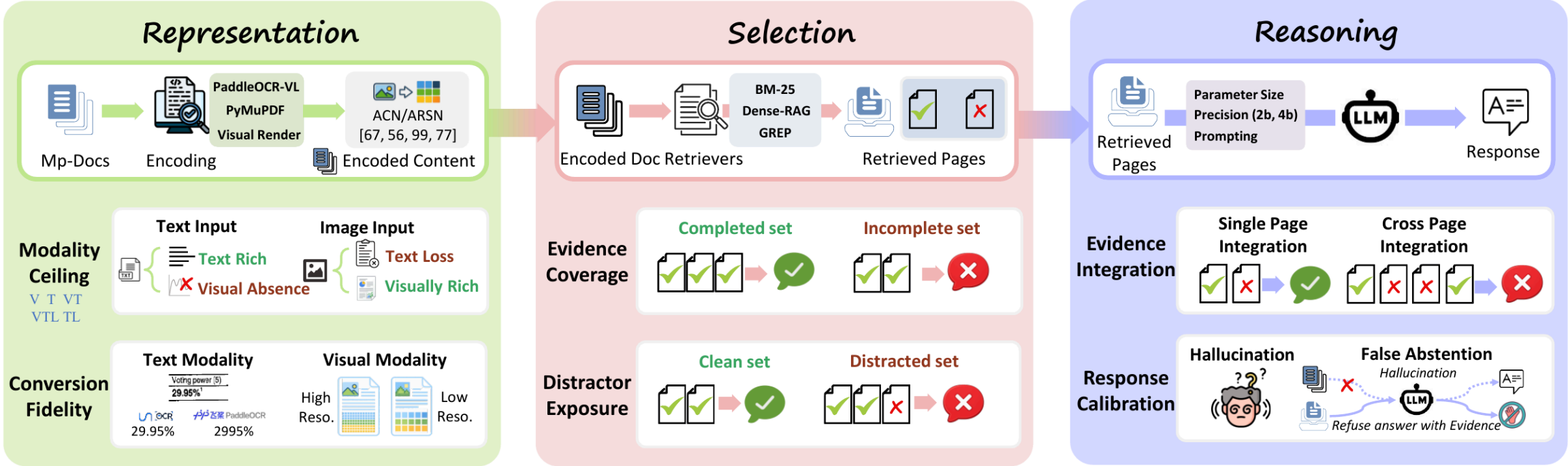
et al., 2025) and that combining text and vision outperforms either modality alone (Han et al., 2025; Suri et al., 2025). A second disagreement concerns **selection**. Fixed-depth retrieval is brittle because a question can require more evidence pages than a fixed depth returns, while too many introduce distractors that dilute attention (Yan et al., 2025; Li et al., 2025). At the same time, other work identifies retrieval coverage as the main factor of accuracy (Liu et al., 2025; Zhu et al., 2025a). A third disagreement concerns **reasoning**. Performance drops on long inputs are attributed to context-length limits (Wang et al., 2024b; Liu et al., 2024), or to failures of evidence integration that persist even when the context is short enough to fit (Zheng et al., 2026; Guo et al., 2026).

- 1 Introduction
- 2 Related Work
- 3 **Attribution Framework**
- 4 Methodology
- 5 **Empirical Findings**
- 6 Discussion and Design Implications
- 7 Deployment Considerations
- 8 Conclusion and Future Work

Submitted to EACL, under review

Bold sections are covered in this talk

Attribution Framework



REPRESENTATION

Was the information preserved when the page was converted?

SELECTION

Did it actually reach the model?

REASONING

Did the model use it correctly?

Modality Ceiling

Evidence	Abstention				Accuracy			
	T	TL	TLV	V	T	TL	TLV	V
Text	23.0	11.5	7.3	8.0	36.9	44.4	51.0	42.3
Layout	20.3	10.2	0.8	4.2	26.3	33.9	49.2	43.2
Table	15.9	7.2	6.7	9.1	35.6	44.7	44.7	34.1
Chart	32.6	28.1	7.3	6.2	16.3	16.9	42.7	40.4
Figure	36.6	24.8	7.6	7.9	14.5	22.1	42.4	39.0
<i>All</i>	26.3	17.2	6.2	7.1	29.0	35.7	49.5	42.6

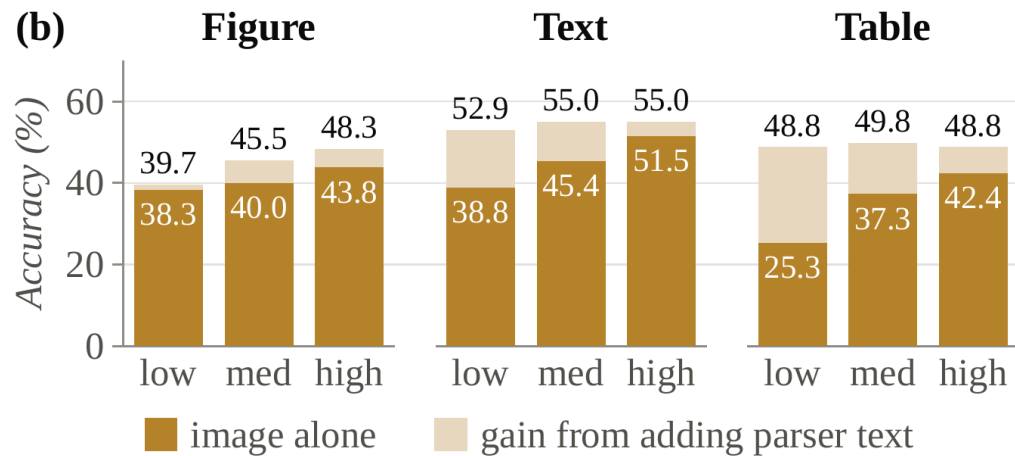
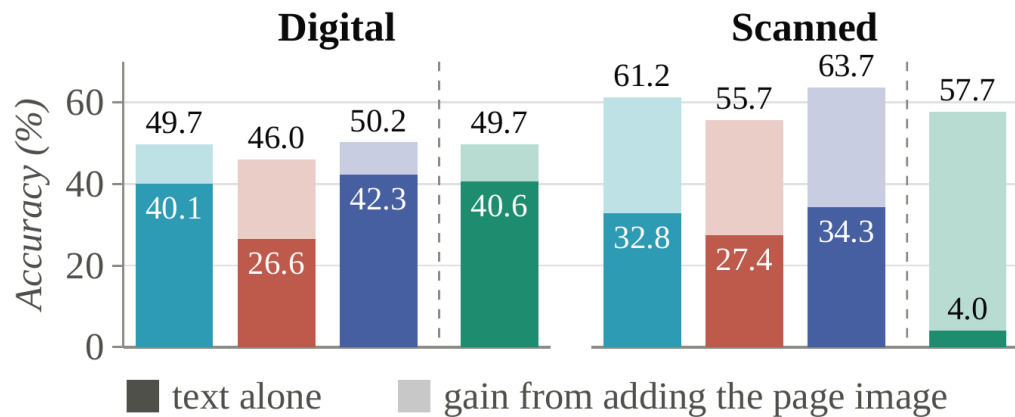
REPRESENTATION

Evidence accessibility depends on modality

- Text-only input refuses **chart and figure** questions most. Adding vision brings abstention **under 10%** everywhere.
- Images alone stay below text plus images: **42.6% against 49.5%.**

Conversion Fidelity

(a) PaddleOCR MinerU Unlimited-OCR PyMuPDF

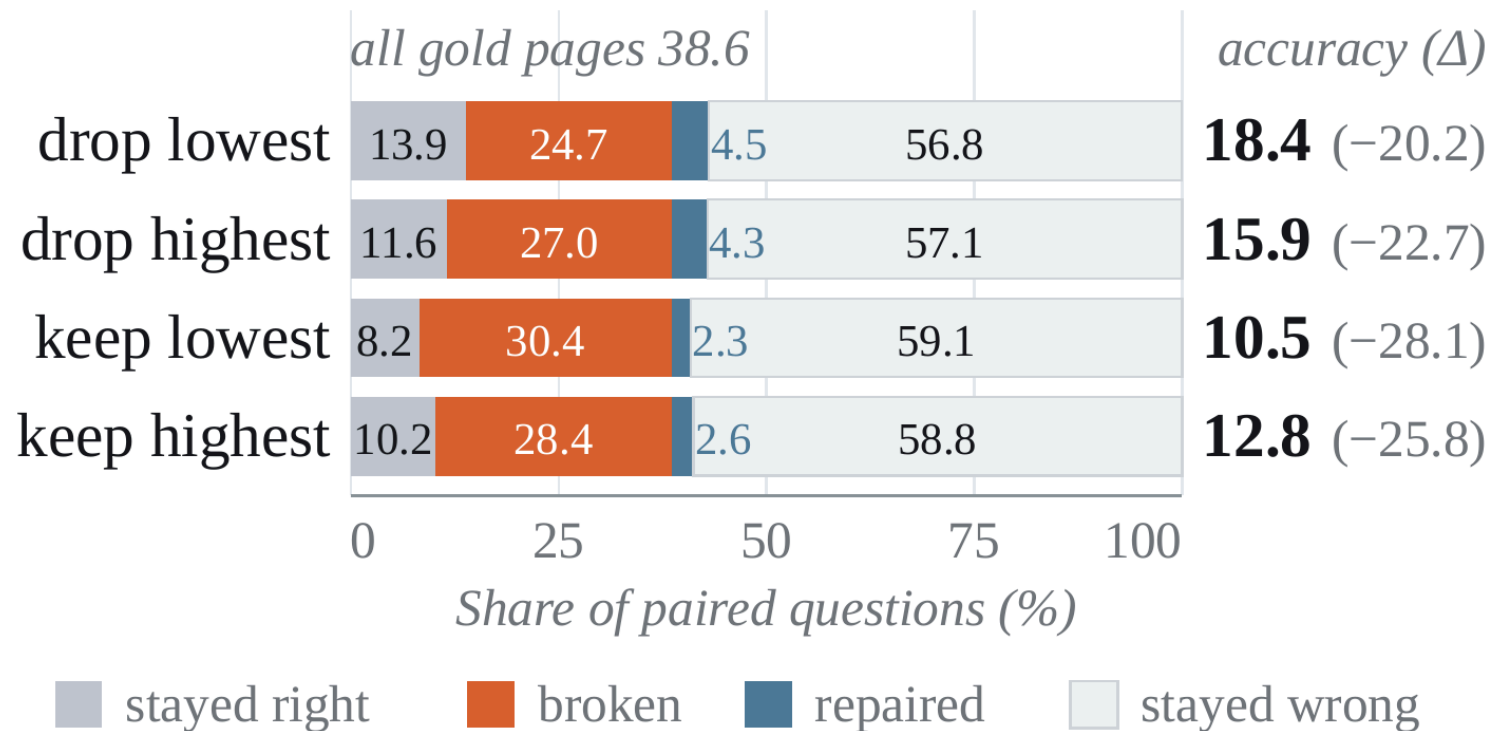


REPRESENTATION

Complementary modalities reduce conversion sensitivity

- The parser spread on digital documents shrinks from **15.7 to 4.2 points** once page images are added.
- Higher resolution helps images alone far more than text plus images.

Evidence Coverage

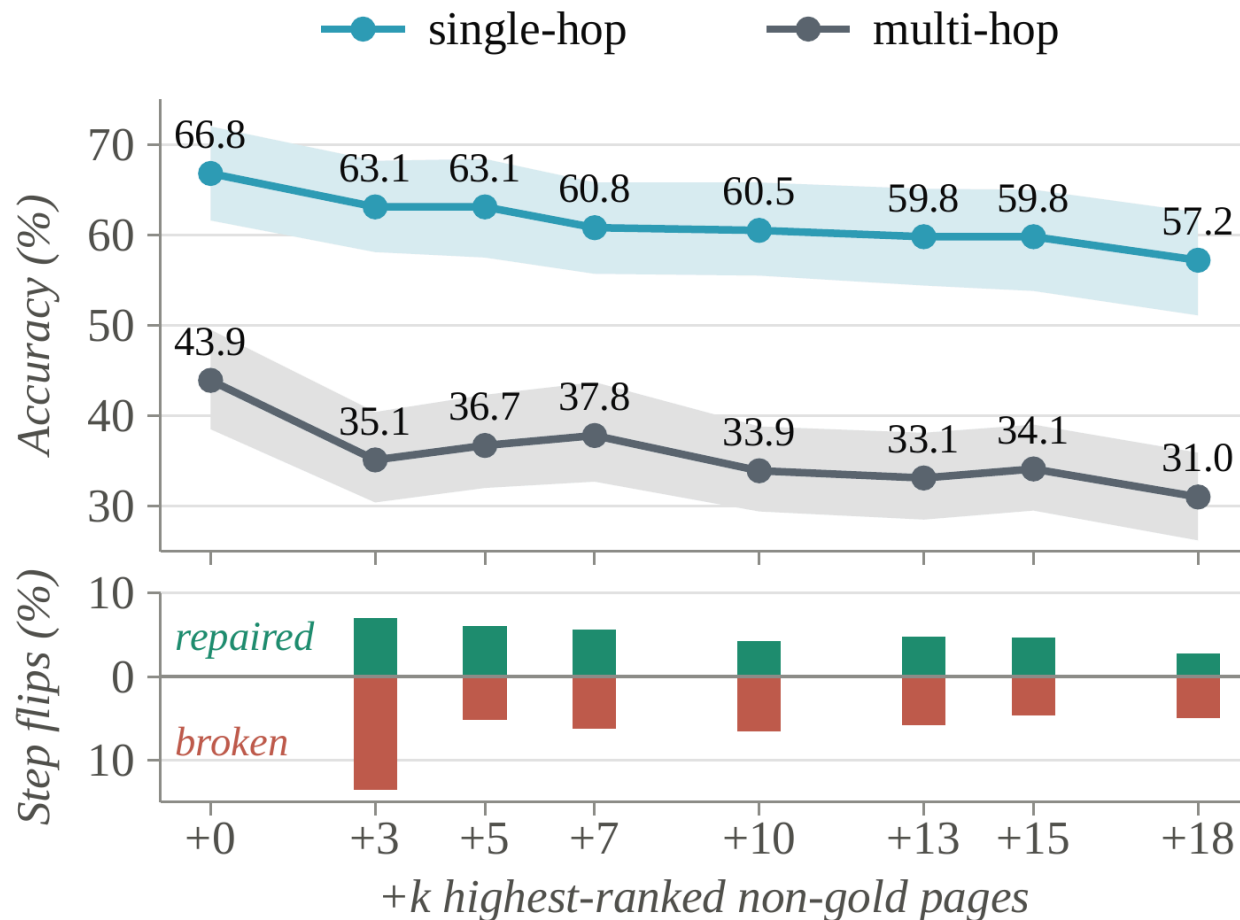


SELECTION

Omission causes large accuracy losses

- Removing one gold page takes accuracy from 38.6% to 18.4% or 15.9%.
- Retrieval rank does not predict which page is safe to lose.

Distractor Exposure

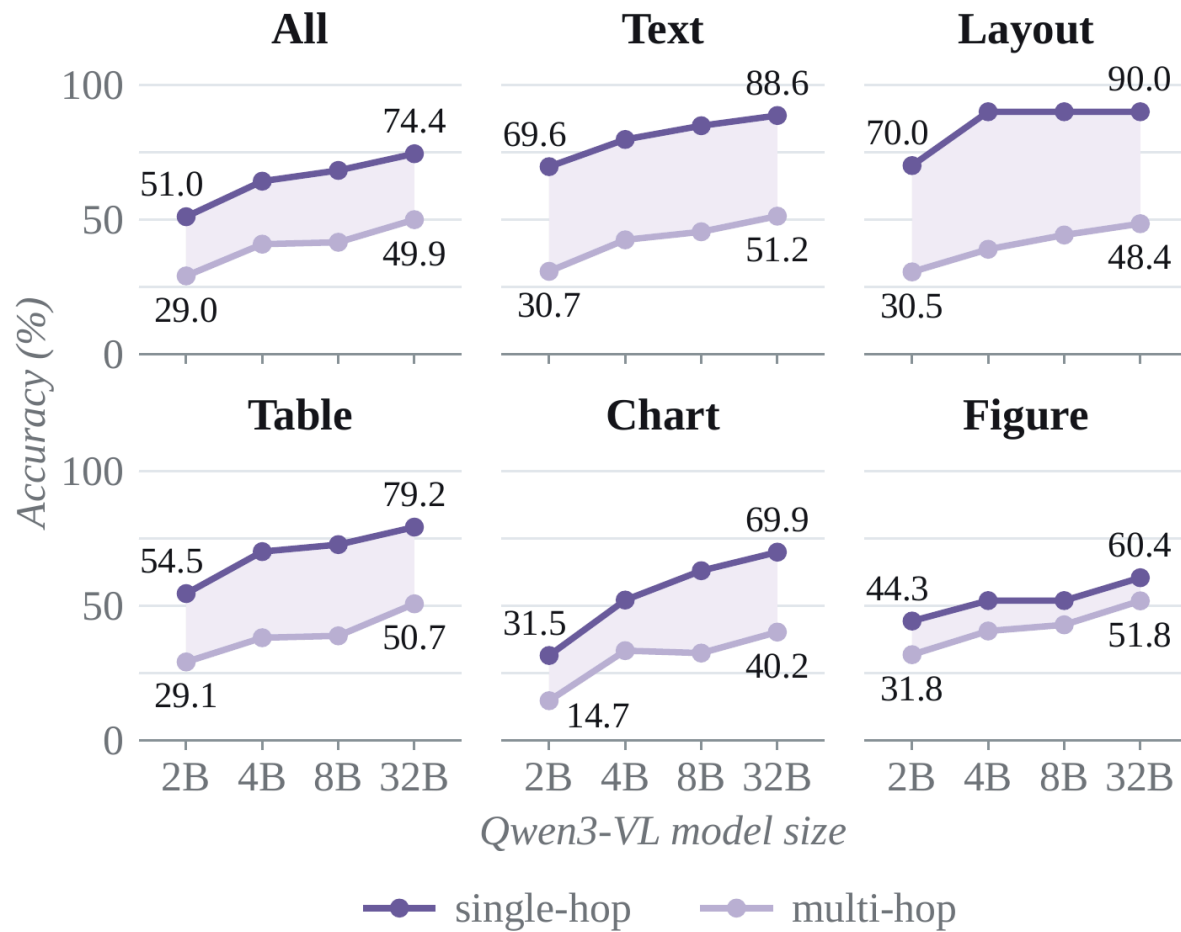


SELECTION

Multi-hop accuracy drops sharply with added context

- Three extra pages: multi-hop falls from **43.9%** to **35.1%**. Single-hop declines gradually.
- The gold pages are all still there. Added pages both break and repair answers.

Evidence Integration

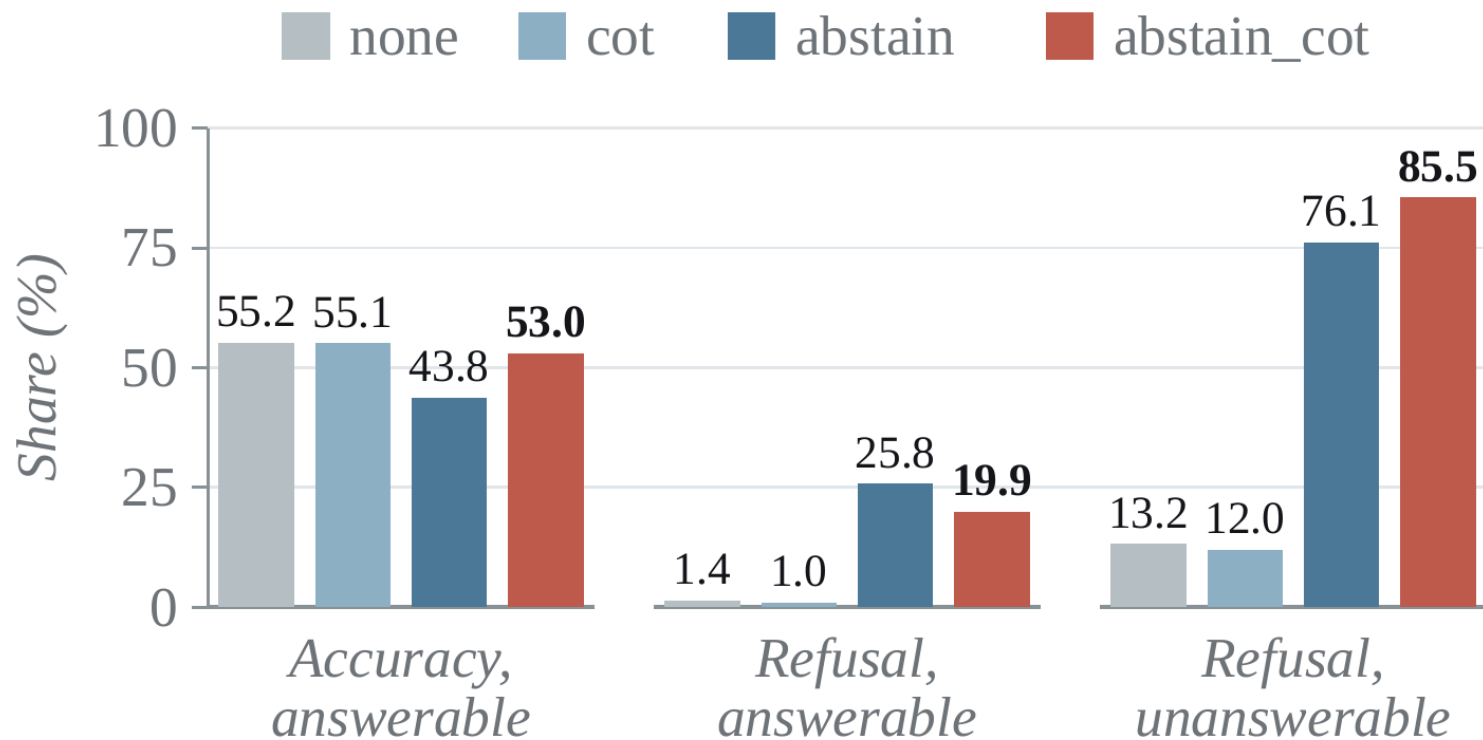


REASONING

Scale improves multi-hop performance

- Gold pages supplied, Qwen3-VL from 2B to 32B: multi-hop rises from **29.0% to 49.9%**.
- Single-hop rises too, so scale helps evidence use in general.

Response Calibration



REASONING

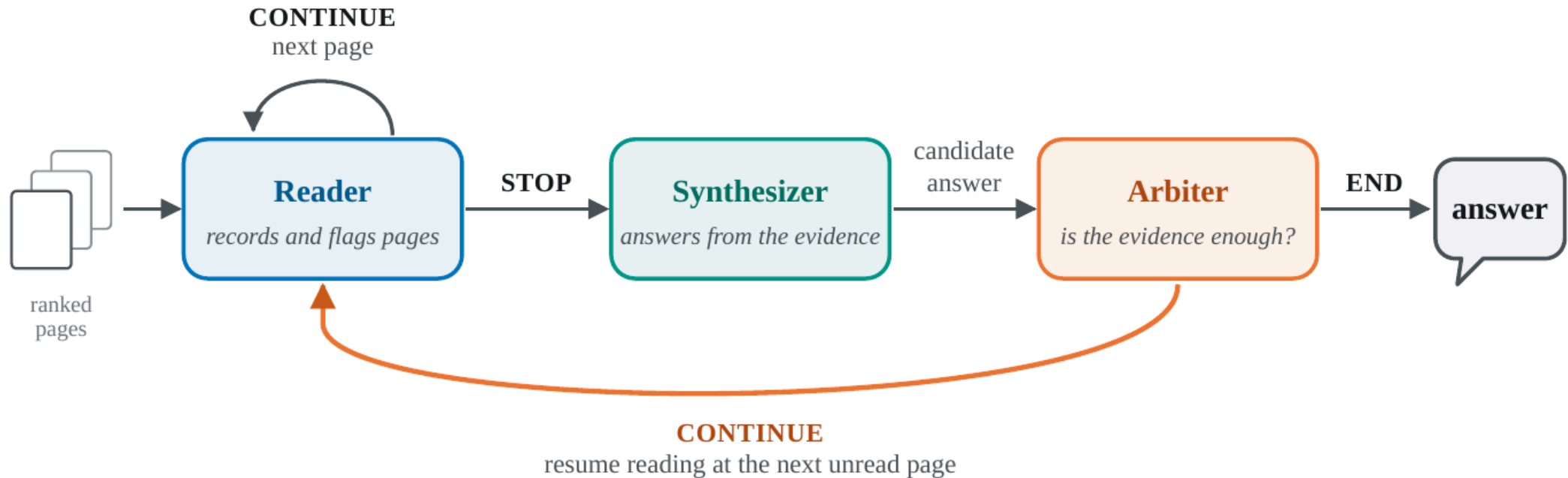
Reasoning before abstaining limits false refusal

- An abstain instruction lifts correct refusal from **13.2%** to **76.1%**, but false refusal goes from **1.4%** to **25.8%**.
- Adding CoT recovers answerable accuracy to 53.0%.

Thank you

Questions?

Simple Method Architecture



READER

Reads pages in retrieval-ranked order, **records evidence**, flags answer pages.

SYNTHESIZER

Answers from the **flagged pages in full** plus the records of the rest.

ARBITER

Ends, or names a missing fact and **sends the reader back**.

Simple Method Results

Method	OCR	Trained	MMLongBench-Doc				LongDocURL			
			SIN	MUL	UNA	ACC	UND	REA	LOC	ACC
Gold pages + VLM	✓	×	57.7	37.4	52.5	49.5	69.7	63.6	59.5	65.7
Whole document (via DocTrace)	×	×	46.9	35.3	25.8	38.5	–	–	–	45.1
M3DocRAG (via SLEUTH)	×	×	–	–	48.6	45.9 ^a	60.1	52.4	45.5	54.6
DocAgent (via MARDoc)	✓	×	–	–	–	46.6	–	–	–	–
MDocAgent (via SLEUTH)	✓	×	–	–	49.7	47.8 ^a	57.5	51.5	44.7	53.1
MoLoRAG (via SLEUTH)	×	×	–	–	51.7	48.8 ^a	63.4	52.7	49.3	57.6
SLEUTH	×	×	–	–	67.4	52.8 ^a	65.7	53.0	53.6	60.0
DocTrace	✓	✓	53.2	41.1	70.5	52.9	–	–	–	56.4
Ours (4-bit)	✓	×	60.5	39.2	76.9	56.5	67.2	66.3	47.9	61.3

- **56.5%** on MMLongBench-Doc and **61.3%** on LongDocURL: highest among the Qwen3-VL-8B systems, training-free, at 4-bit.
- Multi-page questions still trail DocTrace, **39.2 against 41.1**.

Retrieval and Navigation

Retrieval **narrows a long document to compact evidence units** (chunks, pages, or graph nodes) for downstream reasoning.

- **Similarity-based retrieval.** Scores each evidence unit independently against the query and returns the top-ranked subset. Sparse lexical signals such as BM25 are the baseline; dense retrieval uses vision-language late-interaction encoders such as ColPali and ColQwen, often with an LLM reranking stage after.
- **Relation-aware retrieval.** Retrieves along explicit document structure. Tree-based variants follow section hierarchies and outlines, so chunks keep their parent context; graph-based variants traverse typed edges between headings, entities, and cross-references.
- **The trade-off.** Independent scoring handles paraphrase and cross-modal queries, but cannot follow cross-page references or multi-hop dependencies. Relation-aware methods can, at the cost of parser dependence and traversal cost.

Evidence missed or poorly ranked at retrieval is **unrecoverable at generation**. What is retrieved still has to be reasoned over.

Reasoning Strategies

Reasoning strategies act over a **single evidence buffer assembled before the reasoning call**: the full document representation, a retrieved subset, or evidence from an agentic step. None updates model parameters.

- **Sequential reasoning: Chain-of-Thought.** Extends a single reasoning thread. CoT binds evidence from different pages into one chain and decomposes multi-hop questions into single-hop steps.
- **Sequential reasoning: Self-Reflection.** The model evaluates its own draft answer and revises it across rounds, optionally against a verifier or critic.
- **Parallel exploration.** Several reasoning paths over the same buffer. Self-Consistency samples multiple chains and takes the majority answer; sampling-adjudication replaces the vote with an LLM judge. Cost scales with the number of samples. Tree-of-Thoughts has seen little uptake.

A chain can only reason over the evidence already assembled. It **cannot recover what retrieval missed**.

Agentic Methods

Agentic methods put one or more **LLM controllers** in charge: either iterating evidence acquisition under inference-time control, or coordinating across specialised roles.

- **ReAct and tool-augmented reasoning.** A single controller interleaves thought, action, and observation. Actions are document-side tool calls (retrievers, parsers, OCR engines) or structured navigation over a pre-built document representation.
- **Multi-agent orchestration.** Modality-specialised decomposition gives text and visual evidence separate controllers plus a synthesis controller. Role-specialised decomposition splits planning, execution, verification, and synthesis.
- **What the adaptive trajectory buys.** Compute scales with question difficulty rather than document length, the single-step recall ceiling of retrieval-augmented pipelines is lifted, and the action sequence is an inspectable audit trail.

The cost is **controllability**: inference cost is hard to bound, and a controller can commit early to wrong evidence and continue down the wrong trajectory.

Training Strategies

Most systems adopt a backbone **already pretrained on single-page data**, so fine-tuning carries the weight. Training internalises the inference-time evidence-management decisions into model parameters.

Trained retrieval

Modality-alignment aligns visual page embeddings with OCR-derived text. **Cross-page-fusion** adds neighbouring-page context to each page representation. **Logical-relevance** ranks pages by their position in a cross-page reasoning path.

RETRIEVER

Trained reasoning

Trace-imitation supervises the backbone on demonstration chains in a fixed schema. **Reward-driven** training then refines it with GRPO, rewarding answer accuracy, evidence-page recall, and format compliance.

REASONER

Trained agentic methods

Trajectory-distillation clones a stronger teacher’s think-action-observation turns through SFT. **Reinforcement-learned** variants optimise the policy directly against trajectory-level rewards.

CONTROLLER

Two techniques recur across all three: **parameter-efficient tuning** (LoRA, adapters) and **multi-stage curricula** that go from single-page to document-scale supervision. All of it depends on the available datasets.

Datasets

Datasets serve three roles, and the multi-page setting makes all three harder: **per-page annotation does not scale** to documents of tens or hundreds of pages, and long PDFs are harder to source and license than single pages.

Pretraining

The exception rather than the norm. Confined to a few page-to-document and MLLM-centric systems that do continued pretraining on an existing backbone, because multi-page corpora are still small.

RARE

Fine-tuning

Has converged on LVLM-synthesised supervision over scraped PDFs, with reinforcement learning increasingly layered on top to shape evidence-grounding and trajectory behaviour.

LVLM-SYNTHESISED

Benchmarking

Three tiers: single-page extractive baselines, mid-length multi-page suites, and long-document multimodal benchmarks that need cross-page integration. The same small pool of PDFs gets reused.

CONTAMINATION RISK

Answer-only accuracy masks retrieval failure and rewards correct answers reached through incorrect reasoning. Newer benchmarks score retrieval and page-localisation as well, but it is not yet standard.

Experimental Setup

One simple pipeline throughout, with **one condition changed at a time** and everything else held fixed.

Encode

Every page becomes text, an image, or both.
Done once per document and cached, so no question changes what a page contains.

PaddleOCR-VL · 200 DPI

Retrieve

▶ A retriever scores every encoded page against the question and returns the top k , restored to document order.

ColQwen3 · page-level

Answer

▶ A multimodal model receives those pages and the question, and answers in a single pass.
Nothing is fed back.

Qwen3-VL-8B · greedy

- **The stages act once, and in order.** That is what lets a failure be attributed rather than guessed at.
- **Gold pages bypass retrieval.** For most tests the annotated evidence pages are supplied directly, so selection is neutralised and whatever remains is representation or reasoning.
- **Answers are scored by an LLM judge**, Gemini-2.5-Flash at temperature zero. It agrees with GPT-4o on 96.0% and with a human annotator on 98.3% of validation subsets.

Datasets and Evaluation

None of these experiments are possible without annotation that says **which pages support each question**. That is what picks the benchmarks.

MMLongBench-Doc

PRIMARY DIAGNOSTIC

135	47.5	847	244
documents	mean pages	answerable	unanswerable

The only multi-page benchmark we know of that annotates, per question, **the evidence pages, the kind of evidence it draws on, and the document's domain**, while also supplying plausible questions the document cannot answer.

Evidence pages make coverage measurable and define the oracle. Evidence source explains representation results by what the question actually needs. The unanswerable pool makes calibration testable rather than inferred.

Answers are scored by an LLM judge rather than exact match, since a correct response can differ in units, rounding, or formatting. The judge sees only the question, gold answer, and model answer, never the condition being tested. It agrees with GPT-4o on **96.0%** of a stratified sample and with a human annotator on **98.3%**. We track refusals and wrong answers separately, not just accuracy.

LongDocURL

SECONDARY REPLICATION

396	85.6	2,325	8
documents	mean pages	questions	unanswerable

Longer documents, used to check whether the findings hold under a **less demanding setting**. Born-digital throughout, almost no unanswerable questions, and many questions answerable from the text even when the annotated evidence is visual.

Its annotated pages are also not a minimal set: several may be redundant or independently sufficient, which we account for when reading its coverage and integration results.

LongDocURL Transfer

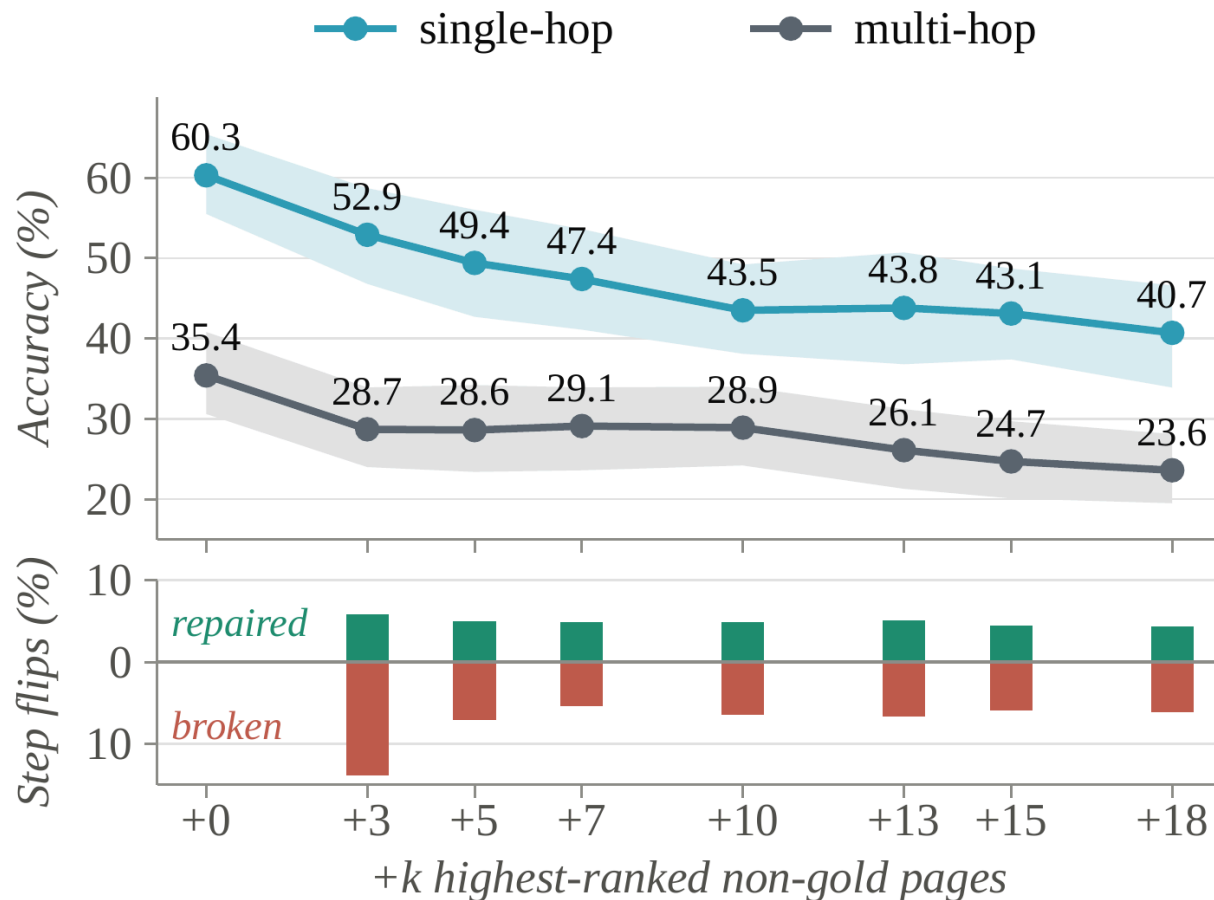
	T	TL	TLV	V
<i>LongDocURL</i>				
Text	75.3	79.5	82.3	74.0
Layout	48.7	52.4	56.3	52.2
Table	61.2	69.5	68.2	57.3
Figure	62.5	65.7	73.4	65.2
All	63.7	68.0	71.9	63.1
<i>MMLongBench-Doc</i>				
Born-digital	40.6	40.1	49.7	41.5
Scanned	4.0	32.8	61.2	58.2

REPRESENTATION

Born-digital documents leave less for vision to add

- Figure questions already reach **65.7%** from parser text. Adding images gives **3.9 points** overall.
- On MMLongBench-Doc, vision adds 28.4 points on scans and 9.6 on born-digital documents.

Distractors on Another Backbone

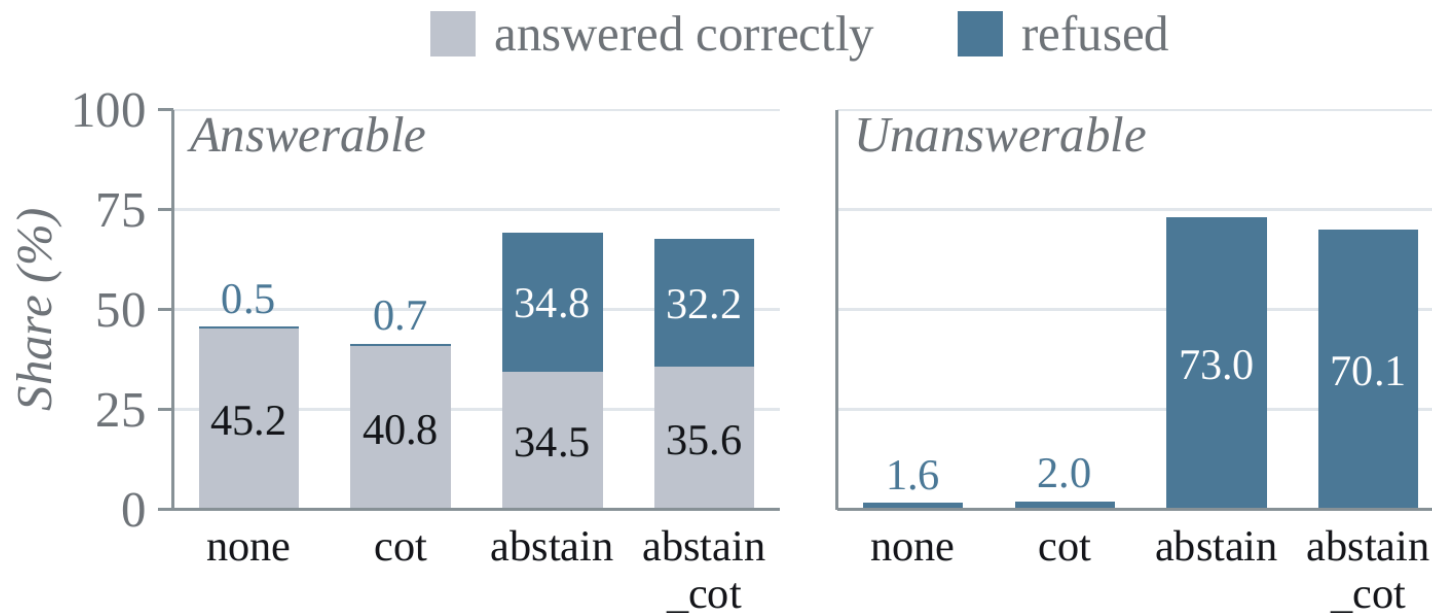


SELECTION

Gemma3-12B shows the same decline

- Single-hop falls from **60.3%** to **40.7%**, multi-hop from **35.4%** to **23.6%**.
- Added context still both repairs and breaks answers.

Calibration on Another Backbone



REASONING

Prompt effects depend on the model

- Adding CoT to abstain gives Gemma3-12B **1.1 points** of answerable accuracy, against 9.2 for Qwen3-VL-8B.
- Correct refusal falls 2.9 points for Gemma and rises 9.4 for Qwen.

Representation Cost

	Acc.	Tokens per page			Latency (s)	
		Text	Vis.	Total	Prefill	Decode
T	29.0	499	0	499	1.4	5.7
TL	35.7	874	0	874	1.8	6.6
TV	49.6	500	459	959	—	—
TLV	49.5	874	459	1333	26.3	6.7
V	42.6	20	459	479	29.0	7.1

DEPLOYMENT

Latency follows visual processing, not token count

- Prefill is **29.0 s** per question with images alone, against 1.4 to 1.8 s for text.
- Embedded text plus images matches parser text plus images: **50.1% against 49.5%**, with no parser.

Reasoner Choice Under Budget

Reasoner	T	TL	TLV	V	GB
<i>Scale (Qwen3-VL, bf16)</i>					
2B	24.7	31.9	38.3	31.9	4.3
4B	32.3	38.6	50.4	42.4	8.9
8B	31.9	38.4	52.4	45.5	17.5
32B	35.5	43.9	59.6	56.7	66.7
<i>Quantized (Qwen3-VL)</i>					
8B, 8-bit	32.5	37.9	53.6	45.7	10.3~
8B, 4-bit	31.8	39.4	51.8	45.5	6.8~
32B, 4-bit	37.0	42.1	59.1	52.3	20.0~
<i>Other families (bf16)</i>					
InternVL3-8B	19.3	25.3	32.6	24.1	15.9
MiniCPM-V-4.5 (8B)	30.7	38.1	60.1	53.3	17.4
GLM-4.6V-Flash (9B)	25.9	32.9	42.7	30.9	20.6
Llama-3.2-11B-Vision	26.5	34.3	33.3	25.3	21.3
Gemma-3-12B	30.5	36.9	48.1	41.6	24.4

DEPLOYMENT

Family and precision matter as well as size

- Qwen3-VL-8B keeps **51.8%** at 4-bit, in about 6.8 GB of weights.
- The 4-bit 32B model reaches **59.1%** in about 20 GB.

From Findings to Design

The survey and the experiments both point at design principles. I put them together in a **deliberately simple system**, to see whether they hold up at the level of a whole pipeline rather than one controlled condition.

REPRESENTATION	Keep text and images together ▶ Every page read at the full TLV representation, with a strong parser and high-resolution rendering, since the two channels cover for each other's failures.
SELECTION	Gather broadly, present selectively ▶ Missing evidence is unrecoverable and moderate distraction is not, so read widely in retrieval-ranked order, but carry most of it forward as compact records rather than as full pages.
REASONING	Check whether the answer is supported ▶ Because the reasoner cannot judge sufficiency on its own, a separate step decides whether to accept the answer or keep reading.

No planner, no tool set, no query reformulation. **The point is to validate the principles, not to introduce a complex architecture.**

Simple Method Cost

	Turns	Tokens			Latency (s)		Peak VRAM (GiB)
		Input text	Input visual	Output	Prefill	Decode	
reader	11.5	5,030	816	697	1.8	46.8	7.2
synthesizer	1.2	9,307	2,018	463	3.8	34.7	8.1
arbiter	1.2	6,806	1,734	464	2.9	33.3	7.9
Per question, mean	13.8	76,669	13,767	9,092	28.5	617.5	8.2
Per question, median	9.0	43,113	9,830	6,078	15.8	385.0	8.0

- **13.8 turns** per question on average, 11.5 of them the reader.
- The cost is time, not memory: **10.8 minutes** mean, 6.7 median, with peak memory at 8.2 GiB.